

AKSHAY R NAIR

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TECHNICAL SUMMARY

Electronics & Communication Engineering (B.Tech) student with hands-on experience building real-time, multithreaded software systems, computer vision pipelines, and backend applications in Python. Strong foundation in data structures, OOP, API integration, and relational databases (MySQL). Seeking a Software Engineering Internship or entry-level Software Engineer role to contribute to scalable, production-grade software.

EDUCATION

College of Engineering Trivandrum B.Tech in Electronics and Communication Engineering	2023 – Present CGPA: 8.22/10
Bhavans Adarsha Vidyalaya, Kakkanad Class XII (CBSE)	2022 94.6%
Bhavans Adarsha Vidyalaya, Kakkanad Class X (CBSE)	2020 90.2%

PROJECTS

Smart Surveillance Checkpost System | *Python, OpenCV, dlib, Multithreading, Raspberry Pi*

- Architected a concurrent, multithreaded Python pipeline separating camera capture, motion-gated face-recognition inference, and asynchronous I/O, eliminating blocking and sustaining a continuous ~28 FPS feed.
- Designed a zero-cloud local architecture with a software-only virtual tripwire (OpenCV frame differencing) to gate compute-heavy inference on genuine motion events, strictly optimizing CPU load for real-time constraints.

Realtime Panoptic Segmentation | *Python, PyTorch, Detectron2, OpenCV, ResNet-50*

- Implemented a Real-Time Object Classification system leveraging a Feature Pyramid Network (FPN) with a ResNet-50 backbone to perform simultaneous instance and semantic segmentation using the COCO dataset.
- Optimized inference performance for CPU portability by implementing an on-demand frame capture mechanism and strict confidence thresholding (≥ 0.6) to filter weak predictions and generate colorized visual masks.

Home Security System using UART | *Embedded C, Python, AT89S52, UART*

- Engineered a two-tier distributed system bridging an embedded sensor node to a PC compute node over a UART serial link, enabling real-time Python-based face recognition and automated local logging.

Nexa AI – AI Placement Mentor | *Python, Streamlit, Google Gemini API*

[Live Link](#)

- Developed a multi-mode AI assistant utilizing Streamlit and Google's Gemini 2.5 Flash API to simulate technical mock interviews, explain core CS concepts, and provide DSA problem-solving guidance.
- Engineered robust API communication with context-aware prompting and automatic retry logic to gracefully handle temporary rate-limit (503) errors, ensuring high system reliability and seamless chat history.

TECHNICAL SKILLS

- Languages:** Python, SQL, C, Embedded C
- Databases:** MySQL
- Concurrency & Systems:** Multithreading (Python), Real-Time Systems, System Design, UART / Serial Comm.
- AI & Frameworks:** PyTorch, Detectron2, OpenCV, Google Gemini API, Streamlit, dlib, NumPy
- Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Concurrency, Hardware/Software Integration
- Tools & OS:** Git, GitHub, Linux (Raspberry Pi OS), API Integration

ACHIEVEMENTS & ACTIVITIES

- Served in the Sea Cadet Corps (Kochi Unit) for four years, advancing to the rank of Leading Cadet while participating in disaster management camps and ceremonial parades.
- Proficient in advanced abacus arithmetic, demonstrating strong mental computation and analytical skills.
- Demonstrated leadership and teamwork through group project coordination at CET.